Project Report: Snake Game Evolved

1. Track Name

Track 1 – (Pygame): Snake with levels

2. Feature List

The project is an implementation of the classic Snake game, built using the Pygame library, with a focus on modular systems and persistent data.

- **Modular Entity System:**
  - **The Snake:** Features advanced movement logic, growth management through a pending_growth queue, and temporary state modifications like speed boosts.
  - **Polymorphic Food System:** A base Food class with multiple specialized subclasses:
    - **NormalFood:** Standard +10 points and +1 growth.
    - **GoldenFood (Big):** High-value (25 points) with significant growth (+5), featuring a time-limited existence.
    - **ShrinkFood:** A strategic item that reduces the snake's length to help navigate high-level stages.
    - **SpeedFood:** Grants a temporary 2× speed and score multiplier for 10 seconds.
- **Dynamic Leveling & Spawning:** A LevelManager tracks the score and increases the game level (up to Level 5). As levels increase, the Spawner unlocks the ability to generate rare food types.
- **Persistent User Profiles:**
  - **Profile Management:** Users create 3-letter profiles (e.g., "AAA") to track their high scores and maximum reached levels.
  - **Food Analytics:** The system records the total count of each food type consumed across all play sessions for every profile.
- **Data Persistence:** Uses a JSON-based storage system (data.json) to save profile progress and maintain a global Top 3 Leaderboard.
- **Hybrid Controls:** Fully navigable menus supporting both Keyboard (Arrow keys/Enter) and Mouse interaction.

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3. Run Steps and User Guide

Setup Instructions

- **Create Virtual Environment:** python -m venv .venv
- **Activate Environment:**
  - **Windows:** .venv\Scripts\activate
  - **macOS/Linux:** source .venv/bin/activate
- **Install Packages:** pip install -r requirements.txt
- **Run Application:** python main.py
- **Freeze Dependencies:** pip freeze > requirements.txt

Menu Navigation & Profile Setup

- **Navigation:** Use the **Up/Down Arrow Keys** to cycle through menu options (Start, Profile, High Scores, Quit).
- **Selection:** Press **Enter** to select an option or click directly on the menu buttons using your **Mouse**
- **Profile Creation:** If "No Profile Selected" appears, navigate to the Profile menu. Type a **3-letter name**. The system will only accept alphabetic characters and will stop at the 3-character limit. Press **Enter** to save the profile to the persistent database.

In-Game Controls

The game uses a responsive control scheme designed to prevent logic errors like "180-degree self-collisions":

- **Movement:**
  - **W:** Move Up
  - **A:** Move Left
  - **S:** Move Down
  - **D:** Move Right
- **Objective:** Collect food to increase your score. As your score hits milestones (50, 100, etc.), the **Level Manager** will automatically scale the difficulty and spawn special food items.

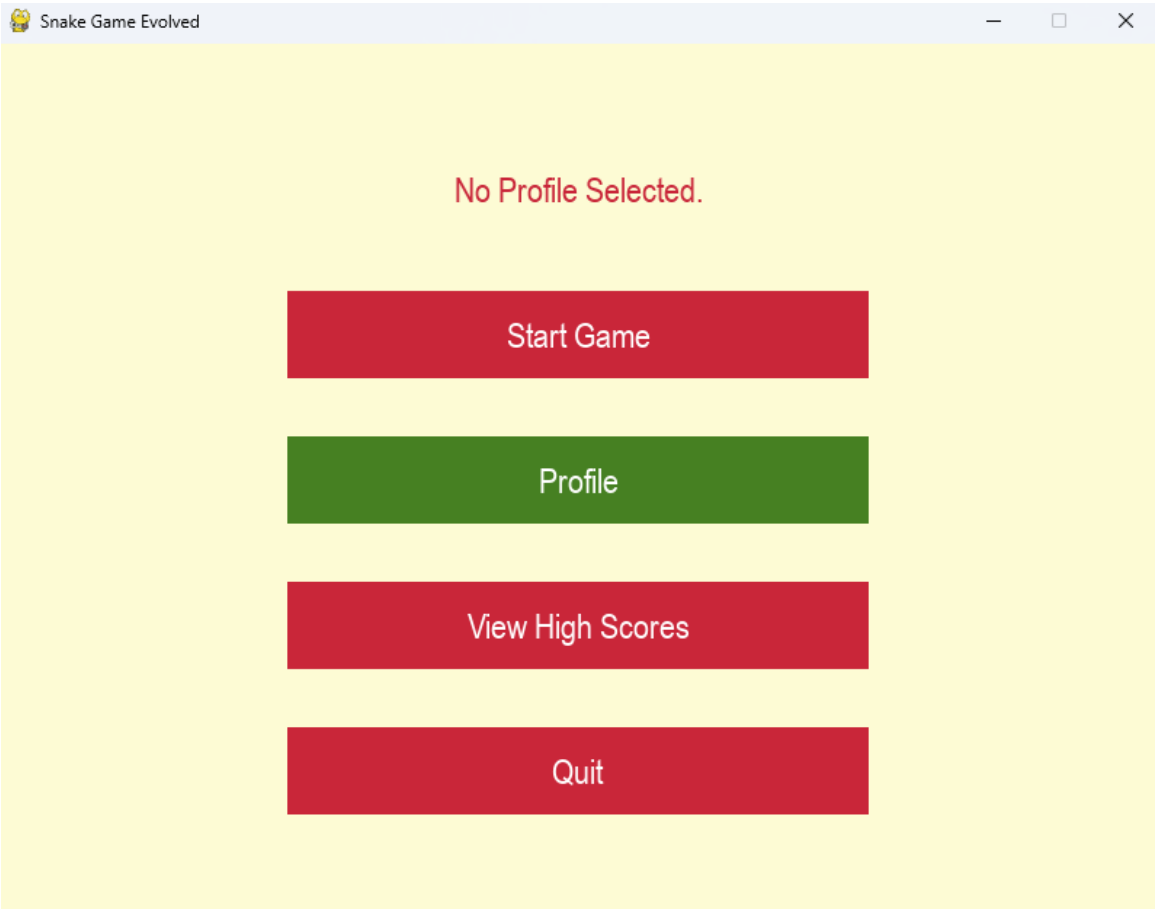
- **Special Items:**
    - **Yellow (Golden):** High points, quick growth.
    - **Blue (Speed):** Temporary 2× speed and score multiplier.
    - **Aqua (Shrink):** Strategically reduces snake length.

Game Over & Results

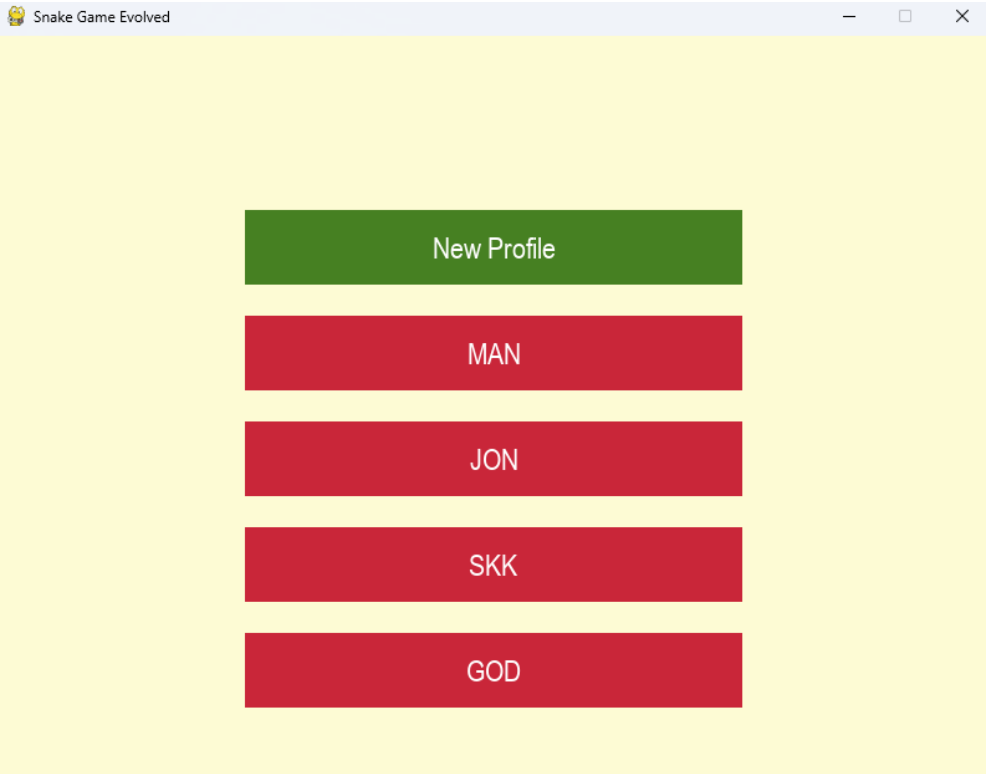
- **Exit to Menu:** Press **ESC** at any time during gameplay or on the Game Over screen to return to the Main Menu.
- **Leaderboard:** After a game over, your results are **parsed and sorted** against existing records. If you reached the Top 3, your name will be highlighted on the leaderboard.

4. Screenshots and Demo Notes

Menu:



Profile Selection and Creation:



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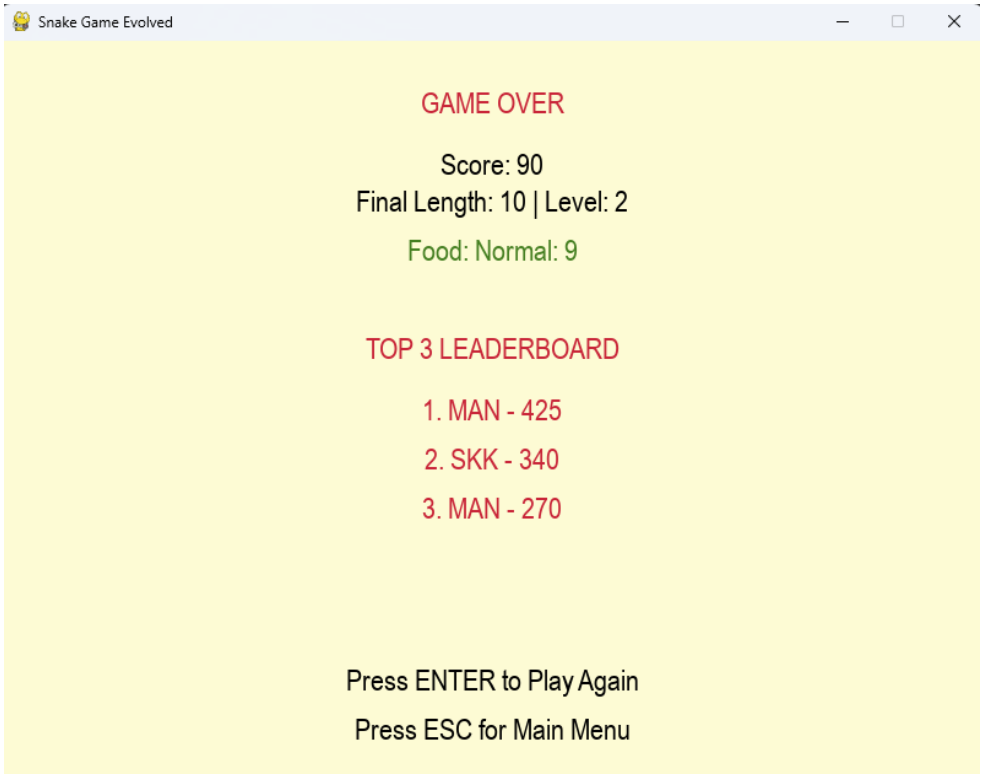
Gameplay:

Game Over:



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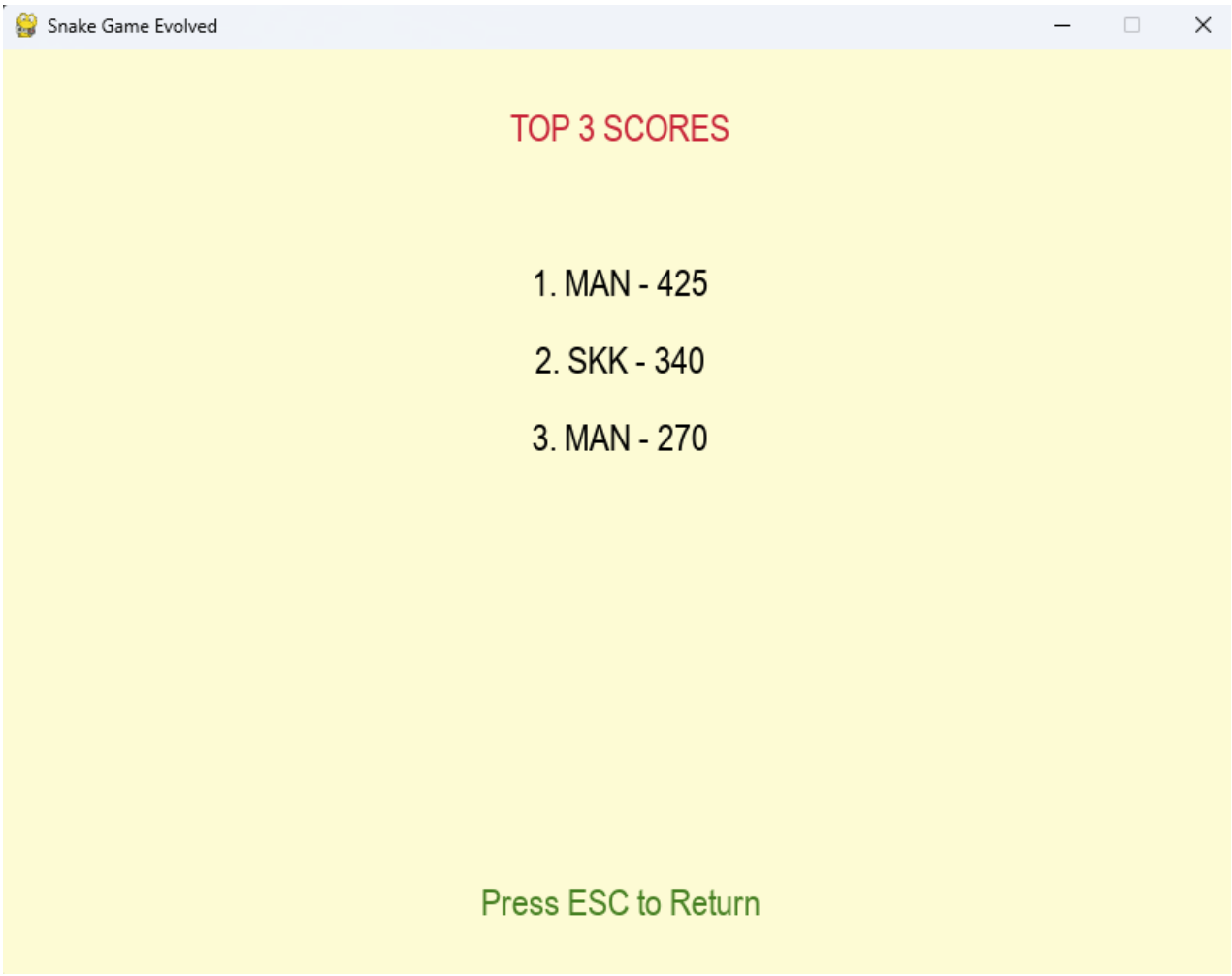
High score screen: